

# Nomad Connect 2024.1

## Release Notes

**7th of May 2024**

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| <b>Issue number</b>    | 2024.1     |
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## Document control

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| Version | Version date | Summary of edits              | Created By | Approved By    |
|---------|--------------|-------------------------------|------------|----------------|
| 1       | 7.05.2024    | First version of the document | Ben Turner | Dieter Termote |

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## What's new – Key changes

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This Nomad Connect release includes:

- Support of RFC8908 Captive Portal API. This allows for smooth logins of modern Apple and Android devices and any other devices that support the RFC.
- Added some features to MAR5:
  - Support for Multi CCU-consists.
  - Possibility to move long sessions (like VPN or CCTV) to best link.
- Telemetry – MQTT, updated to include:
  - Tunnel Status : whether or not connectivity is up or down.
  - Nomad Connect Release: Nomad Connect version + configuration hash.
- Support new Hardware:
  - 3500 with F22P12 and F223-05: facilitate migration of V4 projects to Nomad Connect
  - OBN support for VDS – 28 port 10GBE switch (NMID1143)
- Regular maintenance and security updates : Bug fixes, security patches, avoid need to download Python dependencies, new Wazuh Agent for Nomad Secure,

## Change log – new features

smarter transport, connected passengers

| Key                      | Summary:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">RD-10380</a> | <p>A new solution for dynamic routing was developed, based on the BGP routing protocol. This new solution is more stable and offers several new features.</p> <p>Most importantly:</p> <ul style="list-style-type: none"> <li>• Support for dynamic consists</li> <li>• Support for multi-project CDC</li> <li>• Inter-CDC dynamic routing possible</li> <li>• Route aggregation</li> </ul> <p>This replaces the previous dynamic routing solution based on OSPF, which was created in 2015 and had some inherent issues.</p> <p>While static routing is still possible for projects using MAR3, without DCM, it is advised to upgrade to the dynamic routing when possible.</p> <p>For MAR5 and DCM with failover, this solution is mandatory.</p> |
| <a href="#">RD-10850</a> | <p>All python software now has a virtual environment packaged in the individual debian packages. This means that from 2024.1 onwards, no python dependencies are downloaded at all any more during installation, avoiding the python dependency issues that we had in the past.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <a href="#">RD-9502</a>  | <p>Docker containers can now be deployed to separate docker networks in order to be able to get specific reporting on the volume of data being consumed by individual docker containers. This can be achieved by creating a docker network and then deploying the container in it, telemetry data will then report how much data is consumed by said docker network.</p>                                                                                                                                                                                                                                                                                                                                                                            |
| <a href="#">RD-10751</a> | <p>Features were added to MAR5 in order for sessions to be actively moved between links when necessary.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <a href="#">RD-9826</a>  | <p>From 2024.1 onwards, Nomad Connect now supports the 3500 F22P CCU.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <a href="#">RD-11072</a> | <p>Modifications were made to the captive portal to allow for the timeout of sessions to be modified after they have been created.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <a href="#">RD-10728</a> | <p>Nomad Connect now supports generic facts that can be used with puppet/hiera to create a richer workflow depending on project requirements.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <a href="#">RD-10962</a> | <p>Nomad Connect now supports the RFC8908 Captive Portal API (this is different from our own CPAPI). This allows for smooth logins of modern Apple and Android devices and any other devices that support the RFC.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <a href="#">RD-10993</a> | <p>Nomad Connect telemetry data has been modified to allow the NMS to show the Nomad Connect version that is being used by a CCU in addition to the configuration hash.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

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|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">RD-9849</a> | Nomad Connect will now regularly publish a “tunnel status” message to the MQTT bus, indicating whether the train-to-ground tunnel is up or down. |
| <a href="#">RD-9935</a> | Support for the VDS rail switch of type DS-CSWP28_10GbE_221130-R05 has been added to OBN.                                                        |

## Defects

| Key                      | Summary:                                                                                                                                                                     |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">RD-11040</a> | A bug in the Nomad Connect telemetry code was causing certain messages to be sent twice, this has been resolved.                                                             |
| <a href="#">RD-11027</a> | The latest security updates for Debian12 were incorporated into the release.                                                                                                 |
| <a href="#">RD-10920</a> | Changes were made to the way Nomad Connect mounts media drives to allow for support of disks larger than 2TB.                                                                |
| <a href="#">RD-10887</a> | A bug was causing the upstream throttling of user sessions to fail, this has been resolved.                                                                                  |
| <a href="#">RD-10799</a> | Sometimes gpsd would seem to get stuck on the 4600v3 CCU and cause the last known location to be reported over and over. An upgrade of gpsd to 3.25 has resolved this issue. |
| <a href="#">RD-10796</a> | A bug in the engineering pages was causing modem and other data not to be shown properly on the second CCU in a consist, this has been resolved.                             |
| <a href="#">RD-10769</a> | A bug in the engineering pages was causing GPS data not to be shown properly on the second CCU in a consist, this has been resolved.                                         |

## HW Platforms tested in this release

smarter transport, connected passengers

All formal testing was completed on dedicated QA benches as follows:

| CCU                                                                               | Card / Storage | Modem / mPCIe                                                        | Layer 3                                                |
|-----------------------------------------------------------------------------------|----------------|----------------------------------------------------------------------|--------------------------------------------------------|
| <b>R5001c</b><br>Incl.<br>*A3W10ND1/2<br>(NMID1063/70<br>*A3535ND1<br>(NMID1056)* | SSD            | FN990A40<br>FN990A28<br>EM7565<br>M8 GPS<br>Compex 600<br>Compex 900 | Oring Switch<br>Eltec NAP-1022-2GP-M12X NM31 Switch    |
| <b>4600v3</b>                                                                     | CFast          | FN990A40<br>FN990A28<br>Compex 600<br>Compex 900                     | Oring Switch<br>Eltec NAP-1022-2GP-M12X<br>NM31 Switch |
| <b>4600v2</b>                                                                     | CFast          | EM7455<br>MC7455<br>LM940<br>Onboard GPS<br>Compex 600<br>Compex 900 | Oring Switch<br>Cybox AP1                              |
| <b>R5001B</b><br><br><b>Incl.</b><br>G25A<br>G227                                 | MicroSD        | MC7455<br>LM940<br>M8 GPS<br>7P GPS<br>Compex 600<br>Compex 900      | Eltec Switch<br>Cybox AP2 x2<br>G25A<br>G227           |
| <b>4600v1</b>                                                                     | MicroSD        | MC7455<br>Onboard GPS<br>Compex 600                                  | MEN NL34<br>Cybox AP1                                  |
| <b>B2010</b>                                                                      | CFast          | MC7455<br>EM7455<br>Onboard GPS<br>Compex 600<br>Compex 900          |                                                        |
| <b>R3500</b><br>Incl.<br>SATA Blades<br>G501R<br>F22P12**                         | SSD            | MC7455<br>M8 GPS<br>7P GPS<br>Compex 600<br>Compex 900               | SATA Blades<br>G501R<br>F22P12                         |

\*Testing of A3535ND2 – NMID1057 in 2024-Q2

\*\*Other versions of F22 not supported. The P14 version not compatible with NC. P11 would require further testing.